LAB REPORT

CSE312: Database Management System Lab

Submitted To
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(Lecturer)
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Submitted By
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Experiment No: 02 & 03	Experiment Name: Performing DML and DQL Operations on the national Database Using SQL Queries

1. Experiment Details

In this lab test, the provided **national.sql** database was imported into the local DBMS environment. After importing the database, a set of SQL-based tasks were performed on the region, nation, and orders tables. The tasks included data insertion, update, deletion, filtering with subqueries, and aggregate reporting.

The purpose of this experiment was to develop practical understanding of **DML (Data Manipulation Language)** and **DQL (Data Query Language)** operations using SQL in a real relational database environment.

Assigned Tasks

1. Insert a new region named **South America** in the region table.
2. Update the region name **America** to **North America**.
3. Insert a new nation named **Brazil** in the nation table under **South America** with population **213.5 million**.
4. Delete all records from the orders table where amount is greater than the average amount.
5. Display all nations that belong to the same region as **Bangladesh**.
6. Show the names of nations that belong to the region **Europe** using a subquery.
7. Show all nation records whose population is greater than the average population but less than **213.5 million**.
8. Find the total number of countries in every region and display the result in descending order.

2. Software & Tools Used

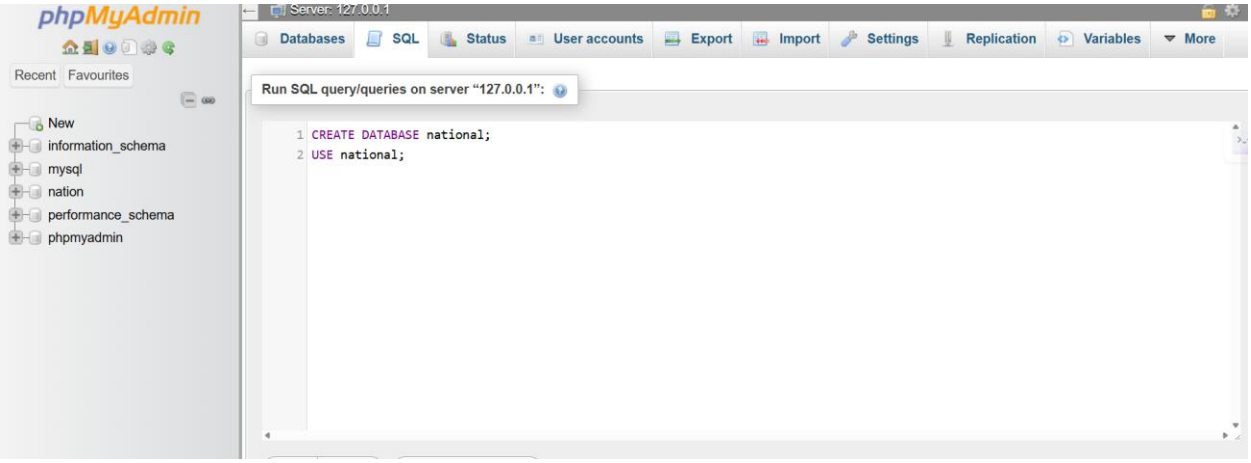
- **Web Server Stack:** XAMPP Control Panel v3.3.0
- **Database Management System:** MariaDB / MySQL
- **Database Interface Tool:** phpMyAdmin
- **SQL File Used:** **national.sql**
- **Text Editor:** Notepad (232-15-818.txt)
- **Operating System:** Windows

3. Methodology

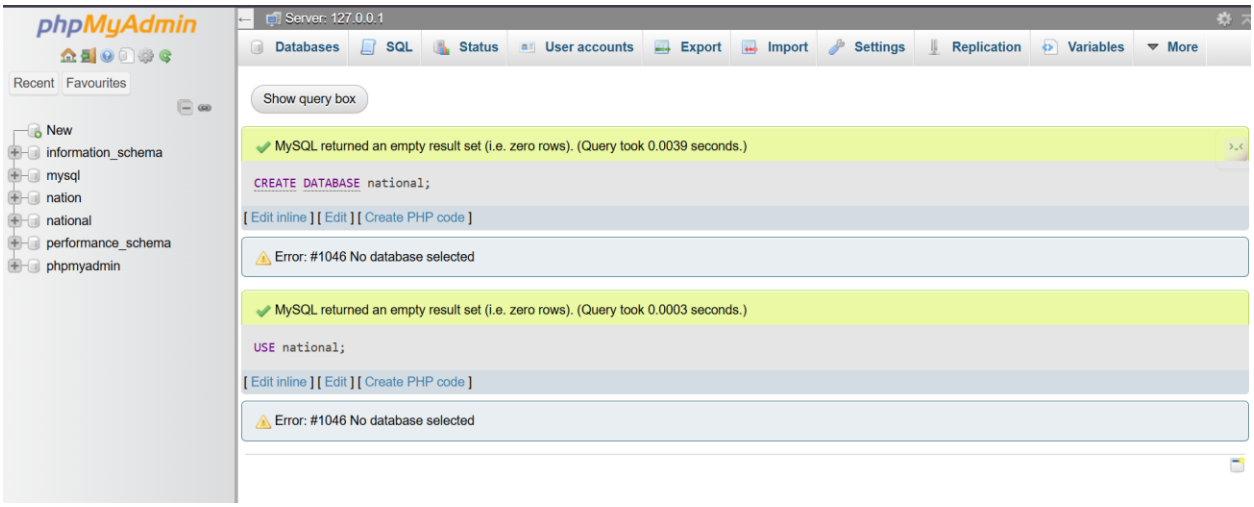
1. The provided **national.sql** file was imported into phpMyAdmin using XAMPP.
2. The required tables such as region, nation, and orders were examined.
3. A text file named 232-15-818.txt was created to store all SQL queries.
4. The given tasks were solved one by one using SQL commands.
5. Each query was executed in phpMyAdmin and the results were verified.
6. For every task, three screenshots were collected: SQL Query, Successful Execution, and Output Result.
7. The outputs were analyzed and then organized into the final lab report format.

4. Experiment Solutions:

Step-01: Create national Database



Step-02: Success of Compilation (national)

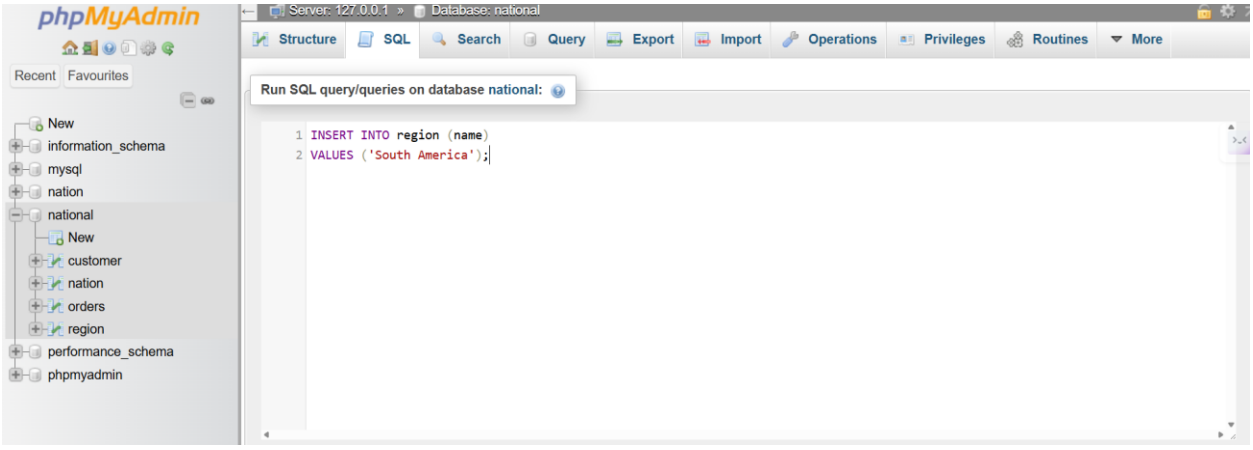


Step-03: Import national.sql file

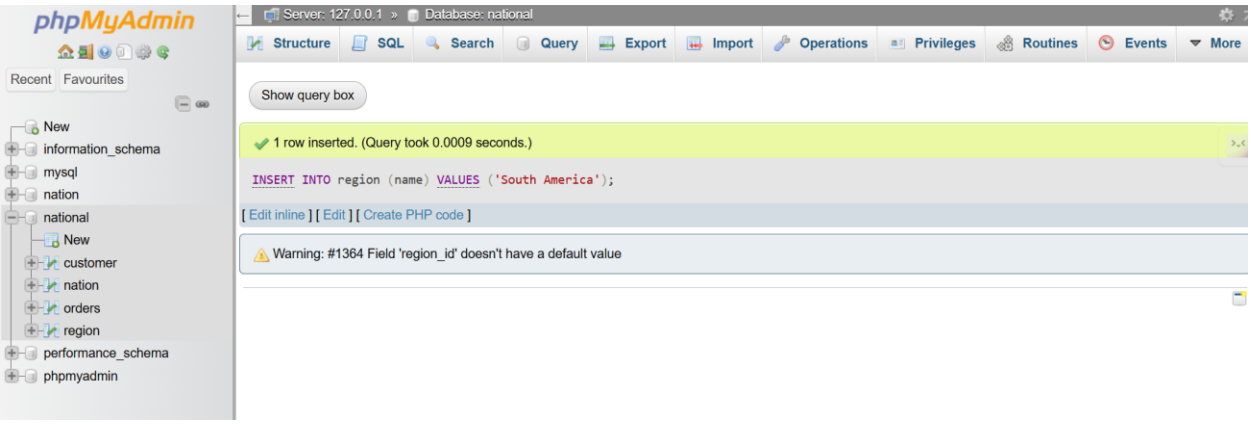
<u>DML PART:</u>

<u>Task 4.1:</u> Insert a new region named "South America" in the region table.

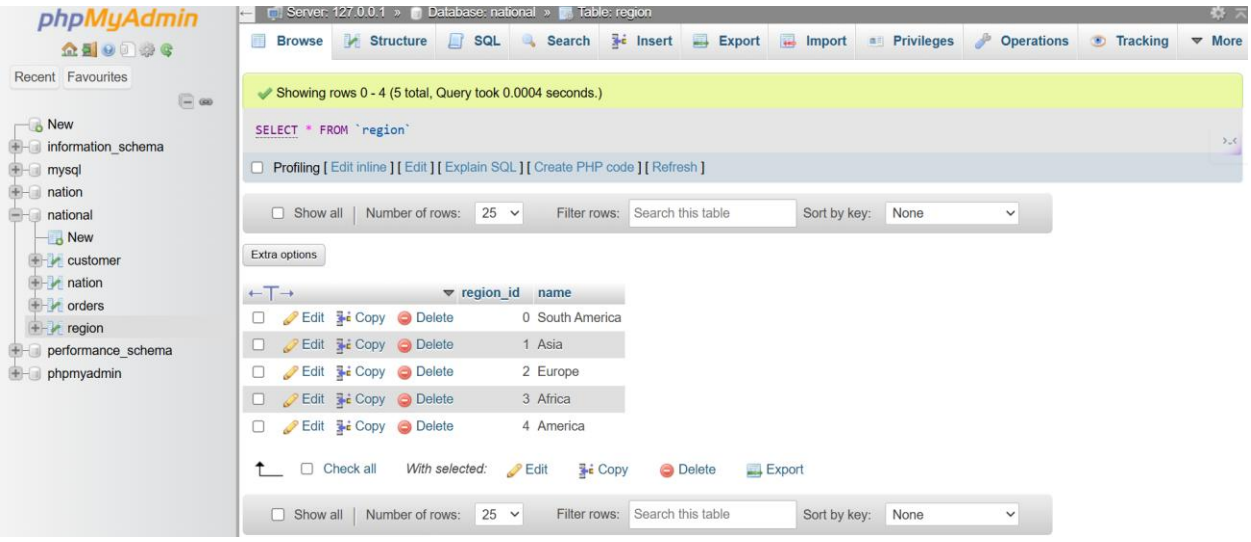
Step-01: SQL Query



Step-02: Success of Compilation

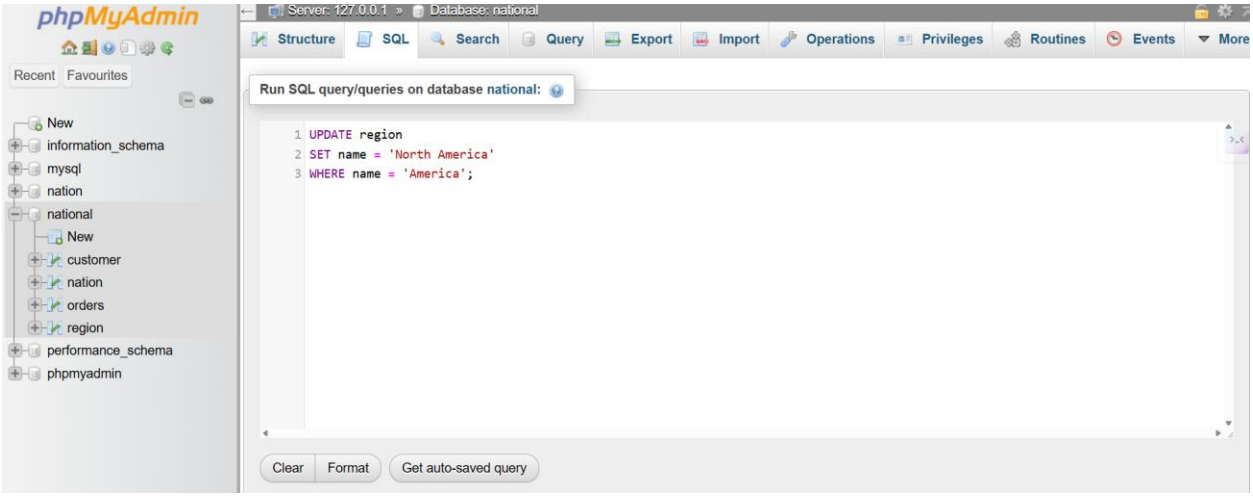


Step-03: Output (region table after insertion)

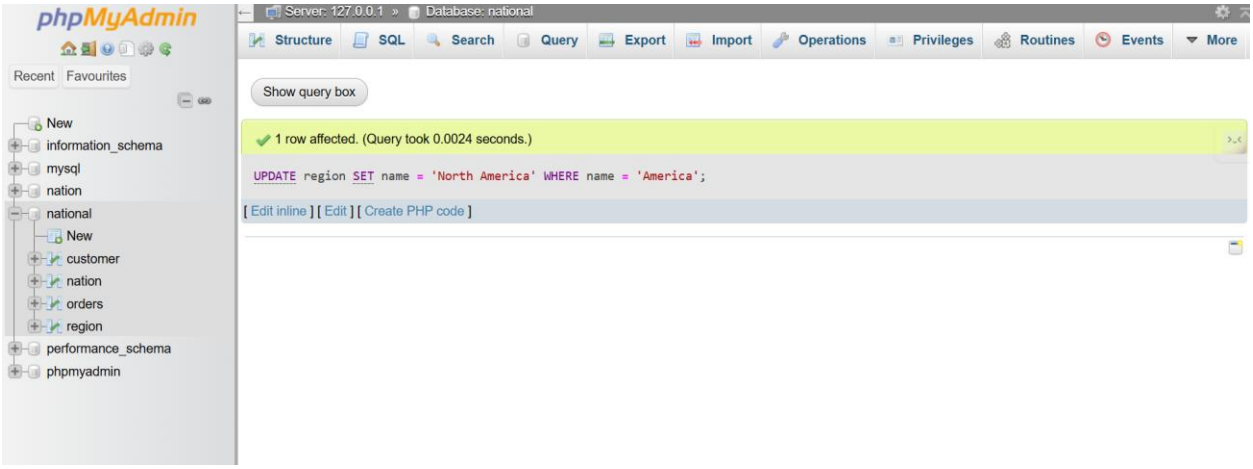


<u>Task 4.2:</u> Update the name of the region named "America" to "North America."

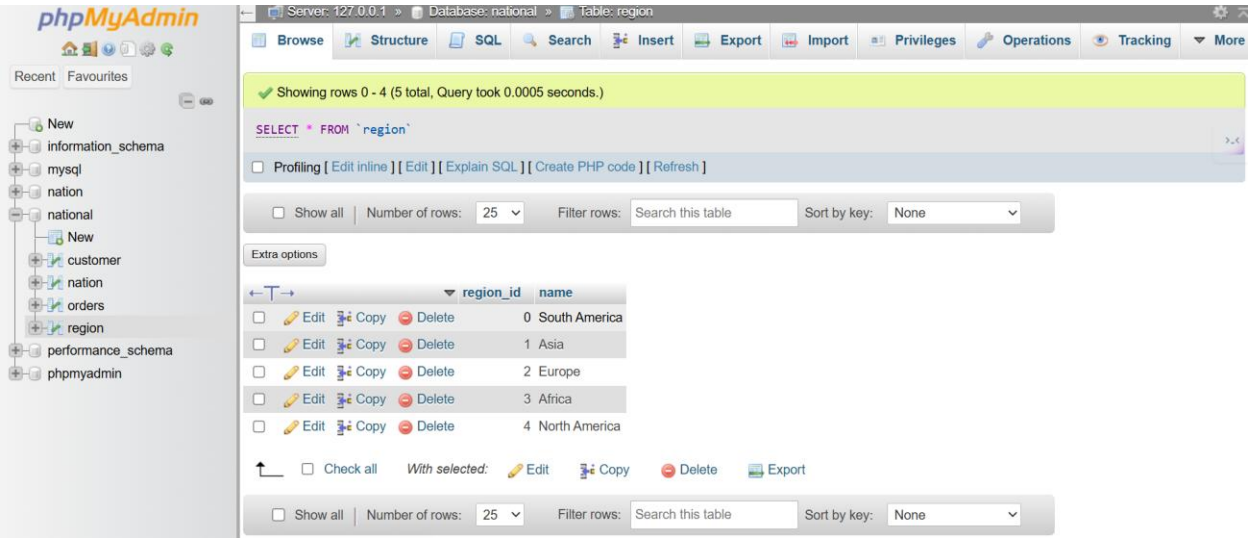
Step-01: SQL Query



Step-02: Success of Compilation

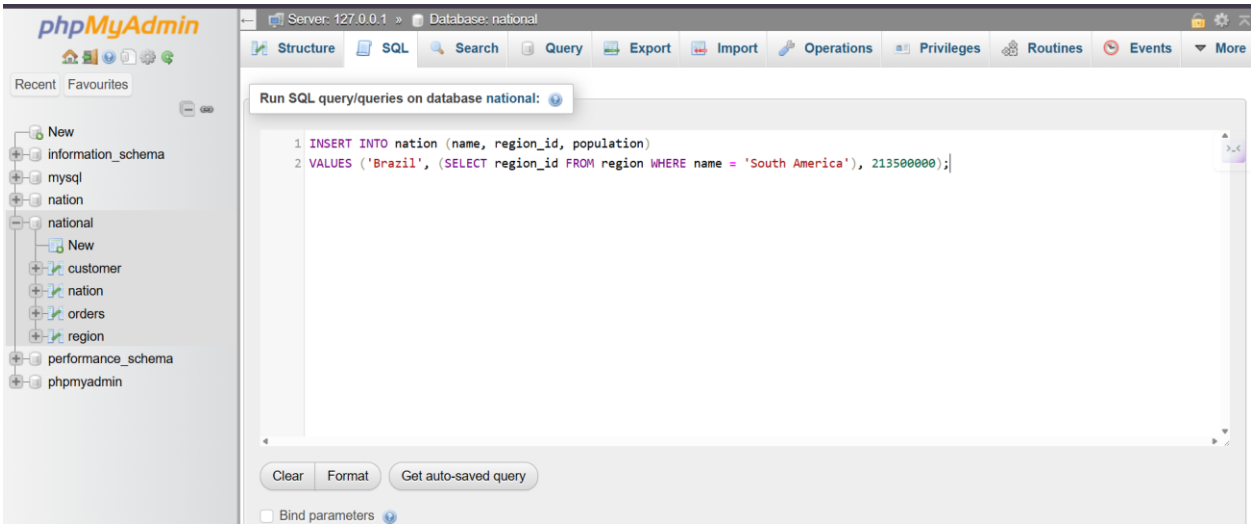


Step-03: Output (region table after update)

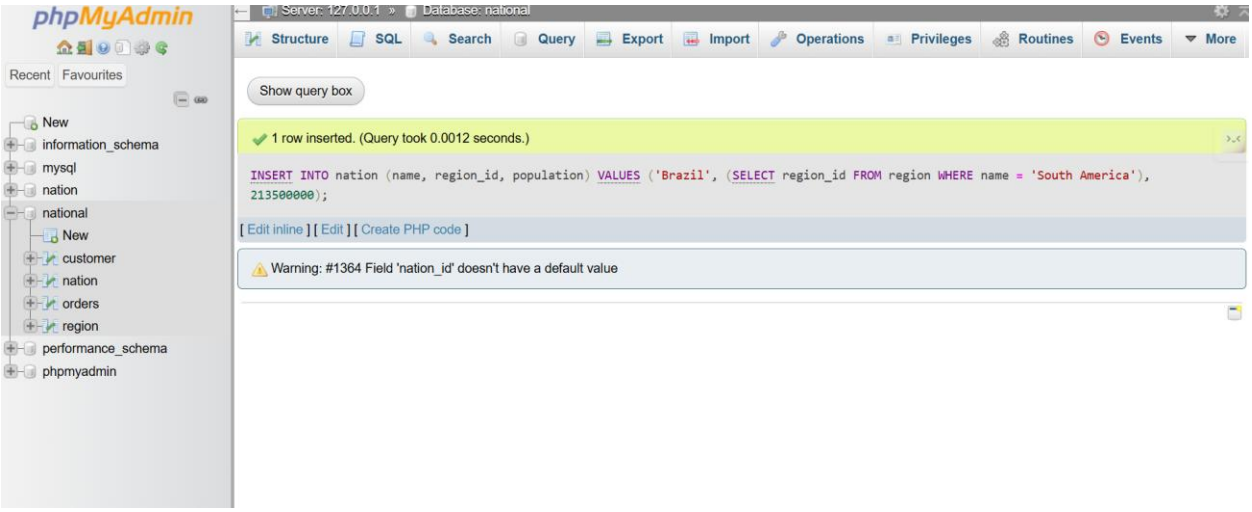


<u>Task 4.3:</u> In the nation table, insert a new entry named "Brazil," which is situated in "South America," with a population of 213.5 million.

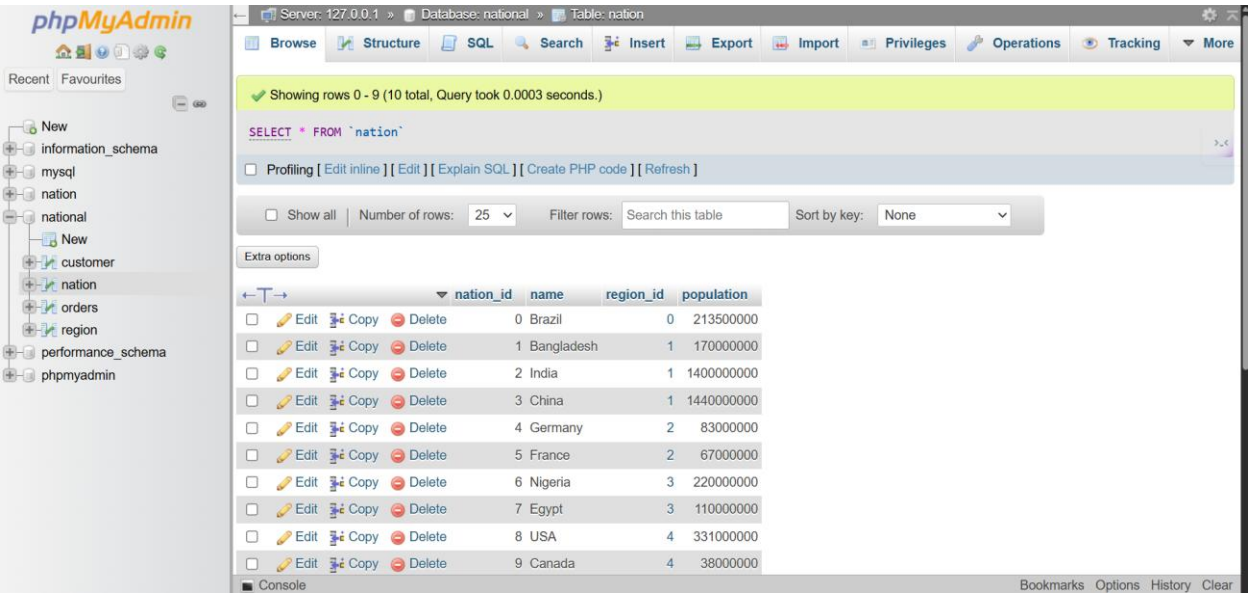
Step-01: SQL Query (Insert into nation)



Step-02: Success of Compilation

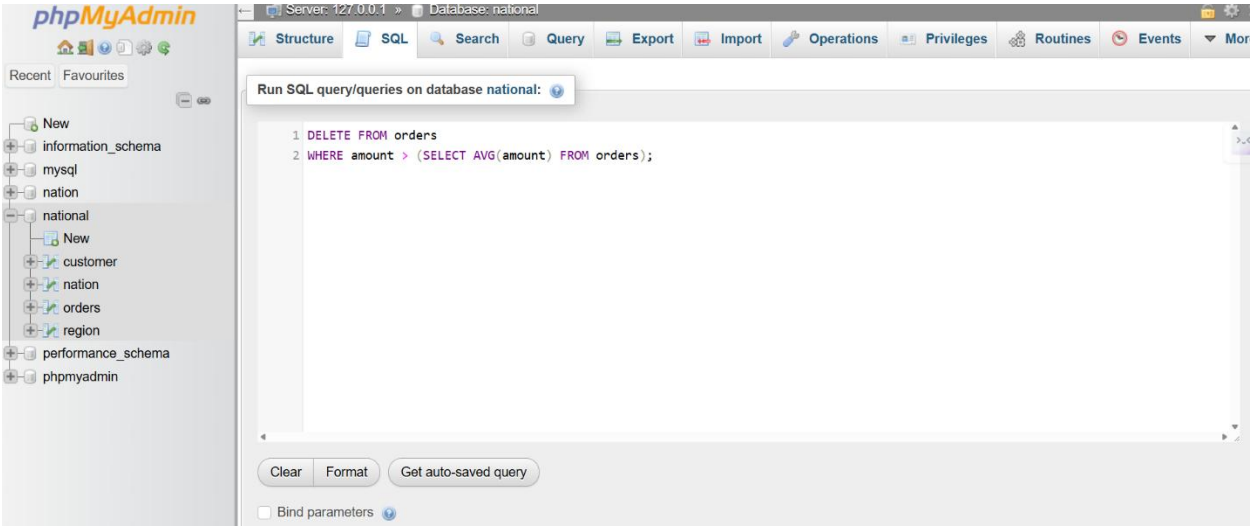


Step-03: Output (nation table after insertion)

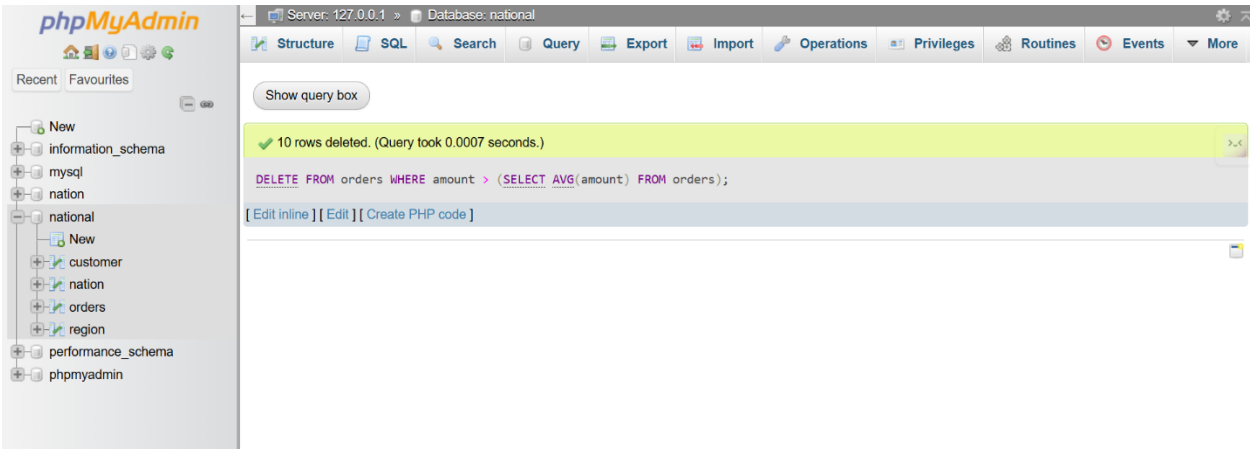


<u>Task 4.4:</u> Delete all the data from the orders table which is more than average order amount.

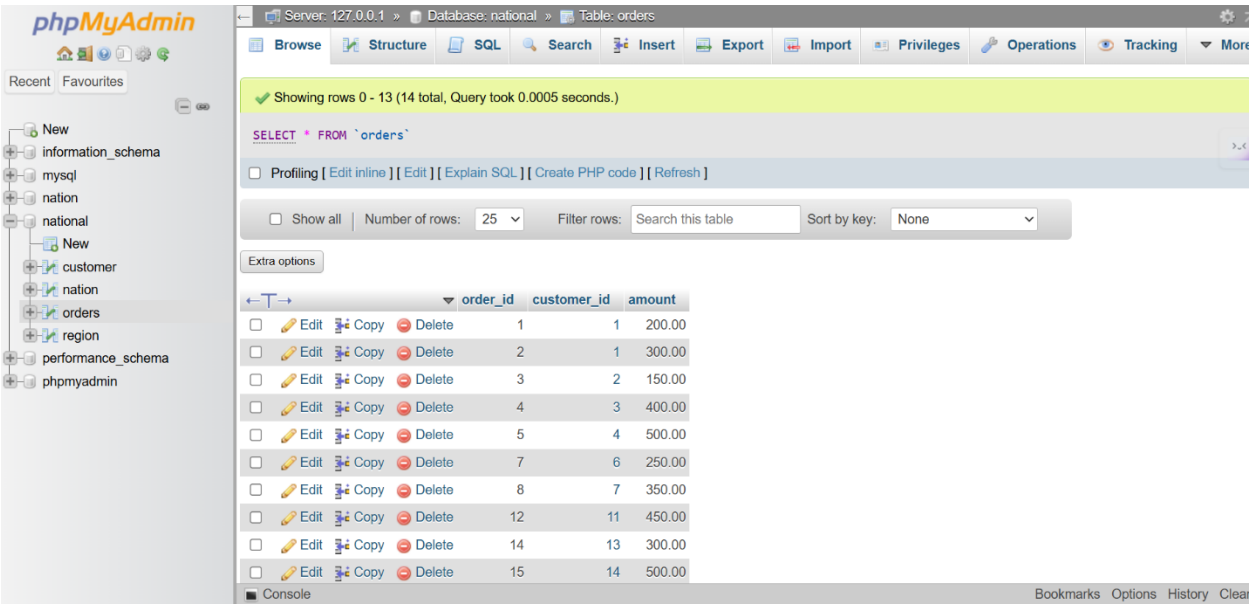
Step-01: SQL Query (Delete from orders)



Step-02: Success of Compilation



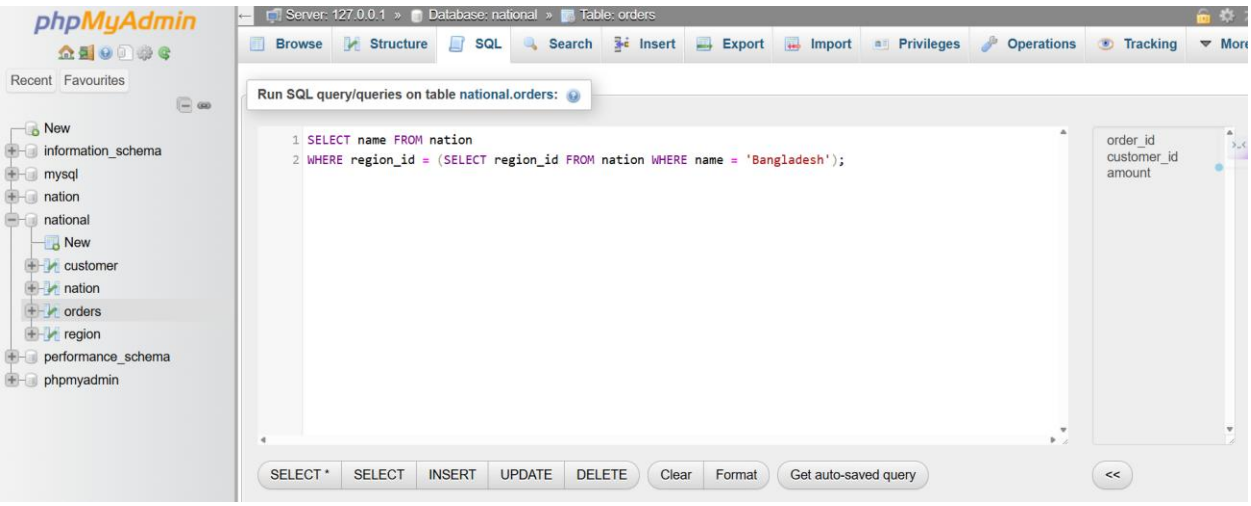
Step-03: Output (orders table after delation)



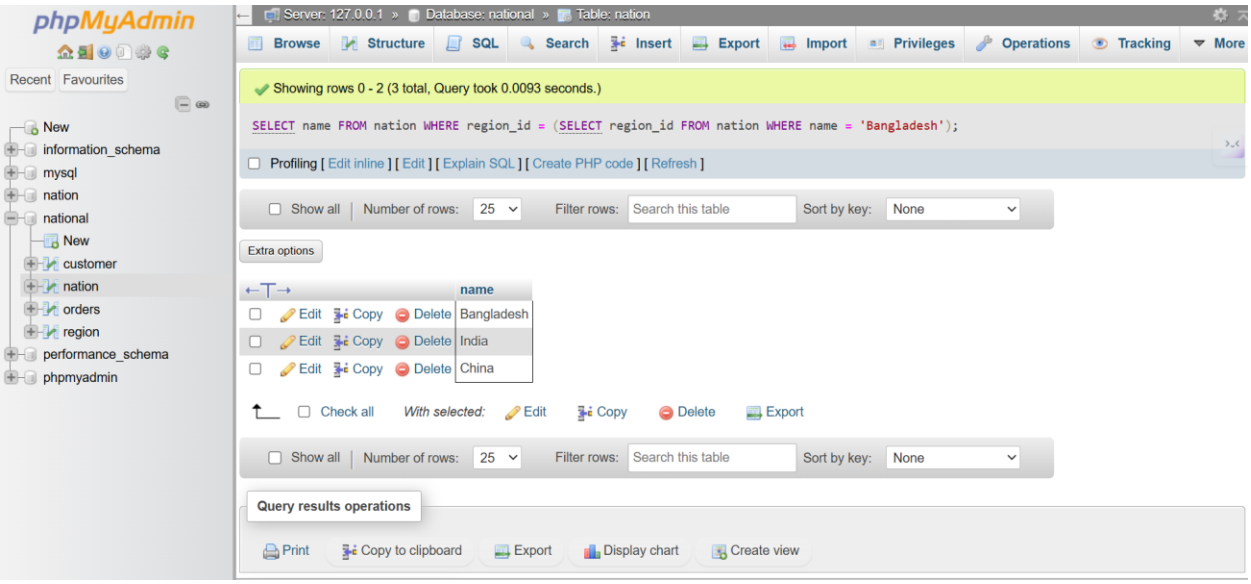
DQL PART:

Task 4.5: Display all nations who belongs to same region as "Bangladesh"

Step-01: SQL Query (same region as Bangladesh)

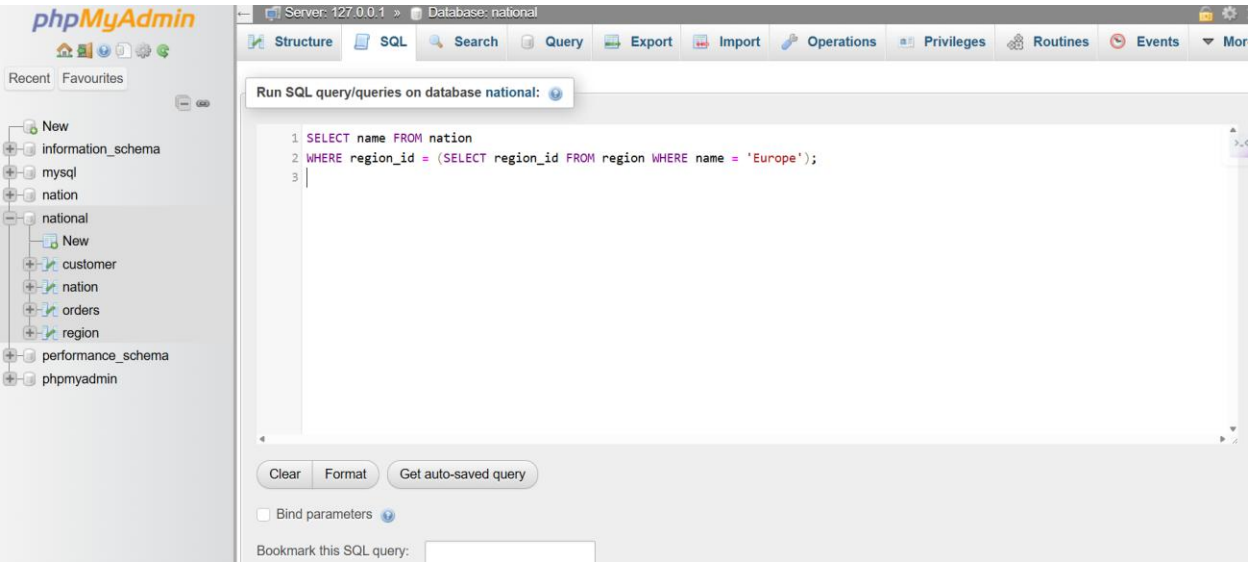


Step-02: Success of Compilation & Output (List of nations)

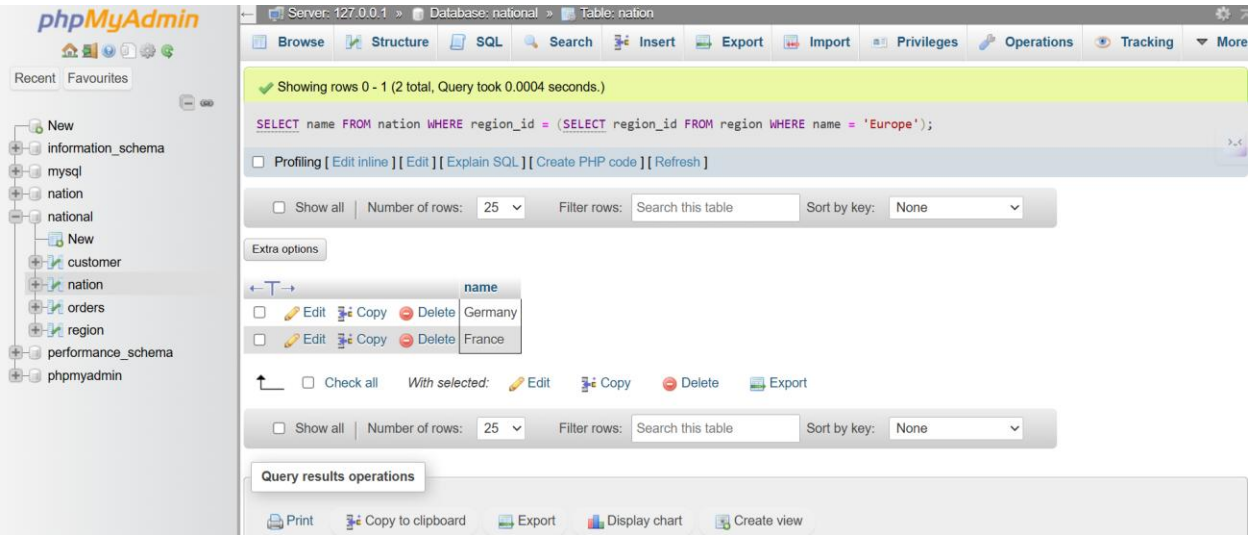


<u>Tasks 4.6:</u> Show the name of the nation that belongs to the region named 'Europe' using a subquery.

Step-01: SQL Query (Nations in Europe)

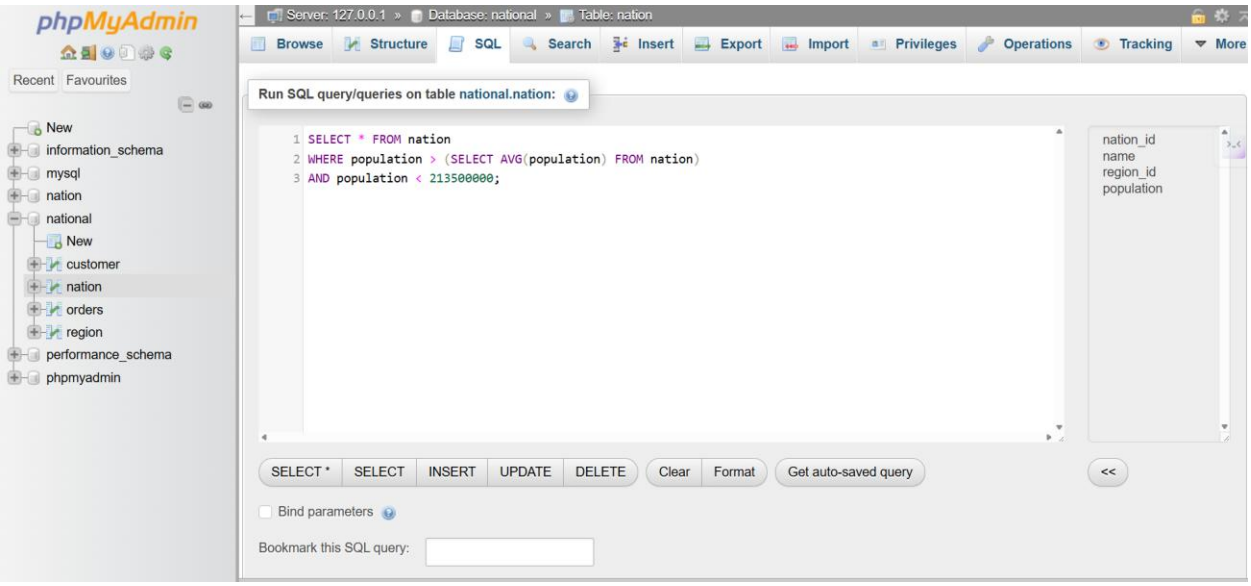


Step-02: Success of Compilation & Output (European nations)

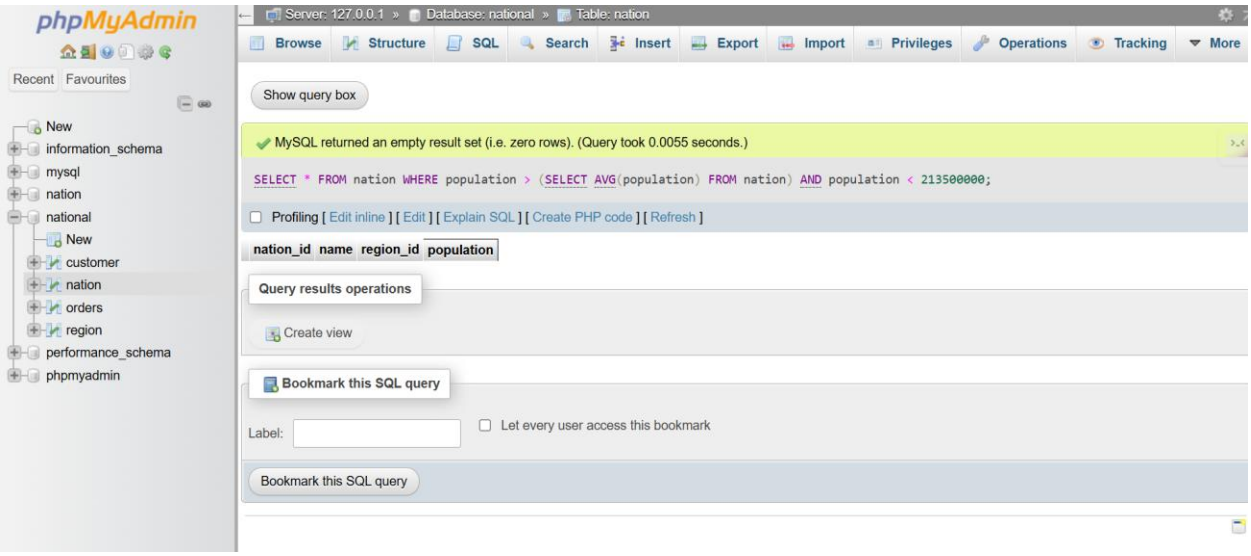


<u>Tasks 4.7:</u> Show all the data of those countries which have higher populations than the average population but lower than 213.5 million.

Step-01: SQL Query

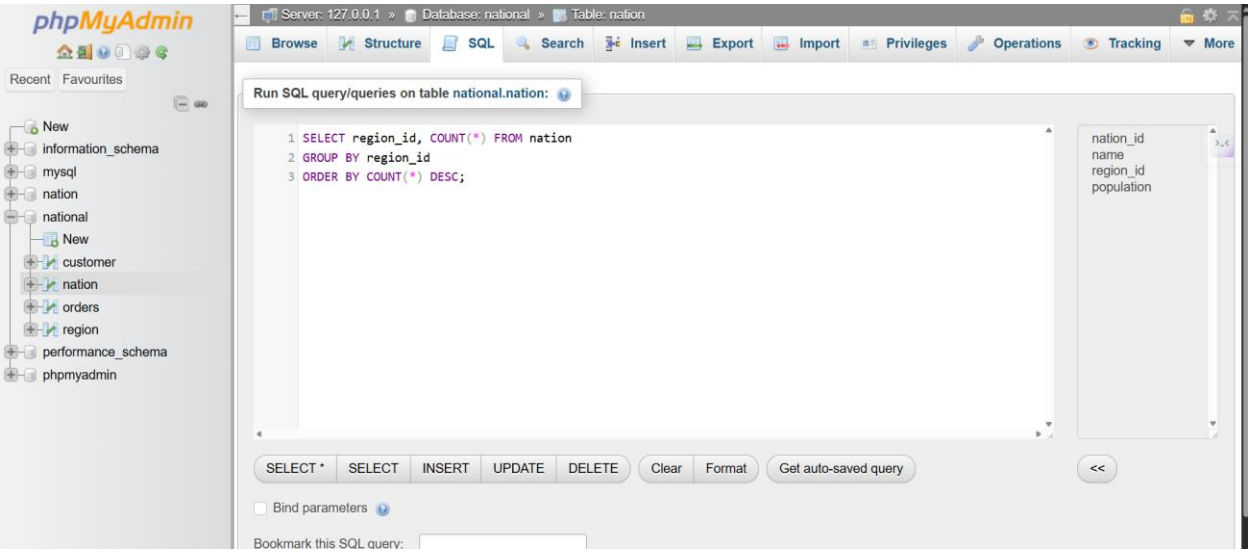


Step-02: Success of Compilation & Output

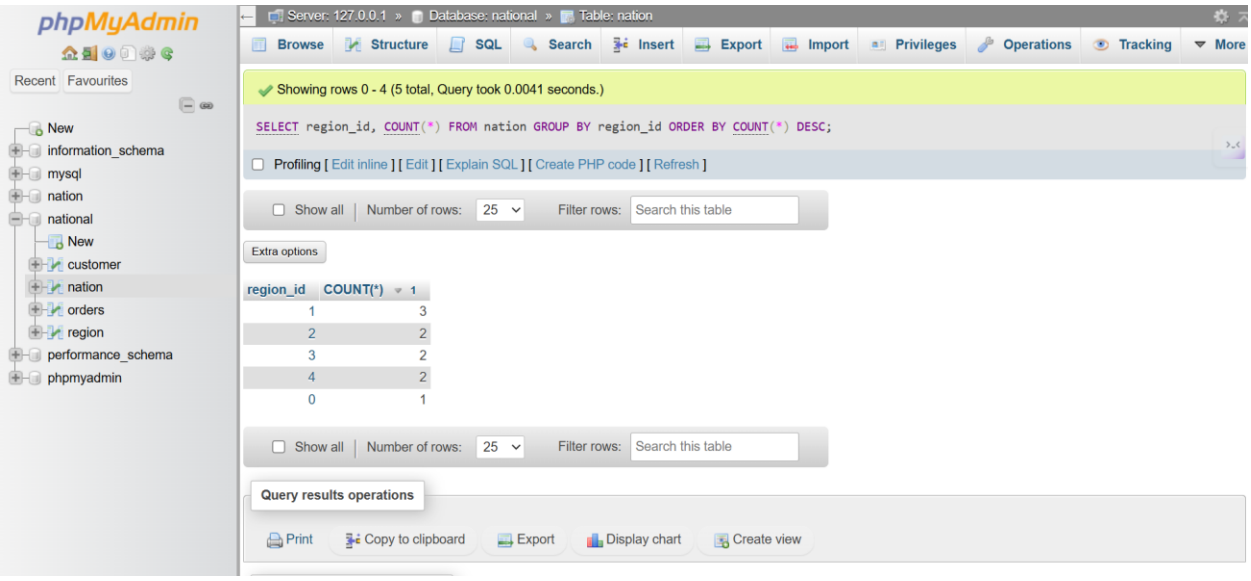


<u>Tasks 4.8:</u> Find the total number of countries in every region in this database and present them in descending order.

Step-01: SQL Query



Step-02: Success of Compilation & Output (region-wise country count)



6. Observation

- The import process of the **national** database was completed successfully.
- DML queries such as **INSERT, UPDATE,** and **DELETE** worked properly on the database tables.
- Subqueries were successfully used to fetch dependent values from related tables.
- Aggregate functions such as **AVG ()** and **COUNT ()** were used effectively for filtering and summarizing data.
- The experiment improved practical understanding of SQL query execution in a relational database environment.
- The report also demonstrated how query results can be made more meaningful by using joins and aliases.

6. Appendix:

Appendix: Course Outcomes, Complex Engineering Problems (EP) and Complex Engineering Activities (EA) Addressing.

Appendix A: Course Outcome (CO) Attainment

SN	COs	Attainment	Justification
01	CO1	Yes	The lab activities developed practical skills in writing and executing SQL queries for database manipulation and retrieval.
02	CO2	Yes	The experiment used relational concepts, subqueries, and aggregate operations on interconnected tables.
03	CO3	No	Advanced topics such as transaction management, security, and distributed databases were not covered in this task.

Appendix B: Addressing Complex Engineering Problems (EP)

SN	Engineering Problem (EP)	Attainment	Related CO	Justification
01	EP1: Depth of Knowledge	Yes	CO1, CO2	The lab required understanding of SQL syntax, query logic, and relational database operations.
02	EP2: Conflicting Requirements	No	CO2	The task was straightforward and did not involve conflicting system constraints.
03	EP4: Familiarity of Issues	Yes	CO1	The experiment addressed standard academic DBMS lab operations.
04	EP7: Interdependence	Yes	CO2	The lab used relations among tables and subqueries dependent on other table values.